

■ 3.3.2 Finding the Structure of a Polynomial

<code>PolynomialQ[expr, x]</code>	test whether <i>expr</i> is a polynomial in <i>x</i>
<code>PolynomialQ[expr, {x₁, x₂, ...}]</code>	test whether <i>expr</i> is a polynomial in the <i>x_i</i>
<code>Variables[poly]</code>	a list of the variables in <i>poly</i>
<code>Length[poly]</code>	the total number of terms in <i>poly</i>
<code>Exponent[poly, x]</code>	the maximum exponent with which <i>x</i> appears in <i>poly</i>
<code>Coefficient[poly, expr]</code>	the coefficient of <i>expr</i> in <i>poly</i>
<code>Coefficient[poly, expr, n]</code>	the coefficient of <i>expr</i> ^{<i>n</i>} in <i>poly</i>
<code>Coefficient[poly, expr, 0]</code>	the term in <i>poly</i> not proportional to <i>expr</i>
<code>CoefficientList[poly, {x₁, x₂, ...}]</code>	generate an array of the coefficients of the <i>x_i</i> in <i>poly</i>

Finding the structure of polynomials written in expanded form.

Here is a polynomial in two variables, written in expanded form.

```
In[1]:= t = Expand[ (1 + x)^3 (1 - y - x)^2 ]
Out[1]= 1 + x - 2 x^2 - 2 x^3 + x^4 + x^5 - 2 y - 4 x y +
        4 x^3 y + 2 x^4 y + y^2 + 3 x^2 y + 3 x^2 y^2 + x^3 y^2
```

`PolynomialQ` reports that *t* is not a polynomial in the single variable *x*.

```
In[2]:= PolynomialQ[t, x]
Out[2]= False
```

t is, however, a polynomial in the two variables *x* and *y*.

```
In[3]:= PolynomialQ[t, {x, y}]
Out[3]= True
```

`Variables` gives a list of the variables in the polynomial.

```
In[4]:= Variables[t]
Out[4]= {x, y}
```

The polynomial *t* has the form of a sum of terms. `Length[t]` gives the number of terms in this sum.

```
In[5]:= Length[t]
Out[5]= 14
```

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This gives the maximum exponent with which x appears in the polynomial t . For a polynomial in one variable, `Exponent` gives the degree of the polynomial.

```
In[6]:= Exponent[t, x]
Out[6]= 5
```

`Coefficient[poly, expr]` gives the total coefficient with which $expr$ appears in $poly$. In this case, the result is a sum of two terms.

```
In[7]:= Coefficient[t, x^2]
Out[7]= -2 + 3 y^2
```

This is equivalent to `Coefficient[t, x^2]`.

```
In[8]:= Coefficient[t, x, 2]
Out[8]= -2 + 3 y^2
```

This picks out the coefficient of x^0 in t .

```
In[9]:= Coefficient[t, x, 0]
Out[9]= 1 - 2 y + y^2
```

`CoefficientList` gives a list of the coefficients of each power of x , starting with x^0 .

```
In[10]:= CoefficientList[1 + 3x^2 + 4x^4, x]
Out[10]= {1, 0, 3, 0, 4}
```

For multivariate polynomials, `CoefficientList` gives an array of the coefficients for each power of each variable.

```
In[11]:= CoefficientList[t, {x, y}]
Out[11]= {{1, -2, 1}, {1, -4, 3}, {-2, 0, 3}, {-2, 4, 1},
          {1, 2, 0}, {1, 0, 0}}
```

It is important to realize that the functions in this section will work as shown above only on polynomials that are given in expanded form. If, for example, you apply `Length` to a factored polynomial, you will get the number of factors, rather than the number of terms in the expanded polynomial.

Many of the functions in this section also work on expressions that are not strictly polynomials.

Without giving specific integer values to a , b and c , this expression cannot strictly be considered a polynomial.

```
In[12]:= x^a + x^b + y^c
Out[12]= x^a + x^b + y^c
```

`Exponent[expr, x]` still gives the maximum exponent of x in $expr$, but here has to write the result in symbolic form.

```
In[13]:= Exponent[%, x]
Out[13]= Max[a, b, 0]
```